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NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Fourth Semester B.E. (Credit System) Degree Examinations
Supplementary Examinations – December 2025

21CS401 / 21IS401 / 21AM401 / 21CC401 – LINEAR ALGEBRA AND ITS APPLICATIONS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Find LU factorization of the matrix A, Where $A = \begin{bmatrix} 2 & 4 & -1 & 5 \\ -4 & -9 & 3 & -8 \\ 2 & 5 & -5 & 1 \\ -6 & 0 & -12 & -3 \end{bmatrix}$	6	L*2	1	1
b)	Find the general solution of following system of linear equations: $x_1 + 2x_2 - 3x_3 - 2x_4 + 4x_5 = 1$ $2x_1 + 5x_2 - 8x_3 - x_4 + 6x_5 = 4$ $x_1 + 4x_2 - 7x_3 + 5x_4 + 2x_5 = 8.$	7	L3	2	2
c)	Using Cayley-Hamilton theorem find the inverse of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 4 \\ -5 & 5 & 6 \end{bmatrix}$.	7	L3	1	1
2. a)	Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -3 \\ -1 & 1 \end{bmatrix}$. Verify that (i) $tr(B^{-1}AB) = tr(A)$ and (ii) $tr(AA^T) = \sum_{i=1}^2 \sum_{j=1}^2 a_{ij}^2$, where a_{ij} is an element in the i^{th} row and the j^{th} column of A.	5	L2	2	1
b)	For what value of k is the following system consistent? $x + y + 2z = 3$ $3x + 4y + 8z = 10$ $-2x + 2y + kz = 7.$	7	L3	1	1
c)	Find the matrix P for the matrix $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ such that $D = P^{-1}AP$ is a diagonal matrix.	8	L3	2	2
3. a)	Row reduce the following matrix to echelon form and locate the pivot position and pivot columns of the matrix A, where $A = \begin{bmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{bmatrix}$	6	L2	1	1
b)	Solve the following system of equations by inverse matrix method $x + y + z = 1$ $4x + 3y - z = 6$ $3x + 3y + 5z = 4.$	7	L3	1	1
c)	Determine the eigen values and corresponding independent eigen vectors of the matrix $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$.	7	L3	2	1

P.T.O.

Unit – II

4. a) Find the dimension of the subspace W of R^3 spanned by the set of vectors $\{(2, 4, 2), (1, -1, 0), (1, 2, 1), (0, 3, 1)\}$. 6 L3 4 1
- b) Determine the matrix of linear transformation $T: R^3 \rightarrow R^2$ defined by $T(x, y, z) = (3x + 2y - 4z, x - 5y + 3z)$ relative to the basis $\beta = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ and $\beta' = \{(1, 3), (2, 5)\}$. 7 L2 3 1
- c) Define the linearly dependent set of vectors. Prove that a set $\{v_1, v_2, v_3, \dots, v_n\}$ of vectors is linearly dependent if and only if at least one of them is a linear combination of other vectors. 7 L2 3 2
5. a) Define the subspace of a vector space. If H and K are subspaces of V , prove that $H \cap K$ is also a subspace of V . 6 L2 3 1
- b) State rank-nullity theorem. Verify rank-nullity theorem for the linear transformation $T: R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x + y, x - y, x + 2z)$. 7 L3 4 2
- c) Find a bases and dimensions of the row space and column space of the matrix, where $A = \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 1 & 4 & 3 & -1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix}$. 7 L2 3 1
6. a) Define the range space of the linear transformation $T: U \rightarrow V$, where U and V are the vector space over same field. Prove that null space of T is a subspace of U . 5 L2 4 1
- b) Define coordinate vector for vector v in the finite dimensional vector space V . Also (i) find coordinate vector of $X = \begin{bmatrix} -3 \\ 7 \end{bmatrix}$ relative to the basis $\beta = \{(1, -4), (-2, 7)\}$. 7 L2 3 2
- (ii) Given $\beta = \{(1, 1), (2, -1)\}$ is a basis and $[X]_\beta = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, find X .
- c) Apply Gram-Schmidt orthogonalization process to the vectors $x_1 = (1, 1, 1, 1)$, $x_2 = (3, 1, 1, 3)$, $x_3 = (5, 0, 2, 3)$ to obtain orthonormal basis for R^3 . 8 L3 4 1

Unit – III

7. a) Find the eigen values of the matrix of the quadratic form $3x_1^2 + 2x_2^2 + x_3^2 + 4x_2x_3 + 4x_1x_2$. 5 L2 5 1
- b) Find a least-squares solution of the inconsistent system $Ax = b$ for $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$. 5 L3 5 1
- c) Find QR factorization of the matrix : $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. 10 L3 5 2
8. a) Define the followings:
- (i) Covariance matrix
- (ii) Total variance of the data.
- (iii) Principal components of the data
- (iv) First principal component. 4 L1 5 2
- b) Three measurements are made on each of four individuals in a random sample from a population. The observation vectors are $X_1 = (1, 2, 1)$, $X_2 = (4, 2, 13)$, $X_3 = (7, 8, 1)$ and $X_4 = (8, 4, 5)$. Compute the sample mean and the covariance matrix. 6 L3 5 2
- c) Construct a singular value decomposition of $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$. 10 L3 5 2